GUIDE 2

Membrane Embedding (CHARMM-GUI)

Goal: Embed a protein-ligand complex into a physiological lipid bilayer and prepare input files for MD simulation.

PHASE 1: File Preparation

1.1. Prepare Complex PDB

Starting from the docking result (Guide 1), the complex PDB must be cleaned and chain IDs fixed before uploading to CHARMM-GUI:

```
# Fix chain ID and atom type for CHARMM-GUI compatibility
sed -i 's/UNL     1/UNL A   1/g' complex_final.pdb
sed -i 's/N1+/N  /g' complex_final.pdb
```

Verify the fix:

```
grep 'UNL' complex_final.pdb | head -3
```

Note: The chain ID (column 22) must not be empty. N1+ atom type causes CGenFF to fail — it must be replaced with N.

1.2. Prepare Ligand mol2 File

CHARMM-GUI requires a mol2 file for ligand force field parameterisation. Use the protonated mol2 from the docking step:

For protonated mexiletine, use pose1_final.mol2 which was verified to contain N.4 atom type.

Note: The mol2 file is used only for atom types and bond orders. Coordinates come from the complex PDB. You do not need to use the docking pose mol2 — any correct mol2 of the same molecule works.

PHASE 2: CHARMM-GUI Membrane Builder (Web)

URL: https://www.charmm-gui.org/?doc=input/membrane.bilayer

2.1. Step 1 — File Upload

- PDB: Upload complex_final.pdb
- Click Next Step

2.2. Step 1 — Chain Selection

- Select ALL chains: ensure all protein chains AND the ligand chain (UNL/HETL) are checked
- In the Ligand Reader section: select 'MOL2 file uploaded from' and upload pose1_final.mol2
- Click Next Step

2.3. Step 2 — Orientation

- Select 'Run PPM 2.0' to automatically orient the protein in the membrane
- Click Next Step

2.4. Step 3 — System Size and Lipid

- Box Type: Rectangular

- Length of XY: 150 Å (start with 150, increase it if 'system smaller than protein' error occurs)
- Lipid Type: POPC (cardiac membrane)
- Ratio: Upperleaflet = 1, Lowerleaflet = 1
- Click 'Calculate Number of Lipids'
- Click Next Step

2.5. Step 4 — Ions and Water

- Method: Monte-Carlo
- Ion Type: NaCl
- Concentration: 0.15 M
- Click Calculate Solvent
- Click Next Step

2.6. Step 5 — Assembly

- Click Next Step to assemble components
- Wait 15–25 minutes for lipid/water placement

2.7. Step 6 — Download

- Input Generation Options: check OpenMM
- Click Download .tgz
- Save the Job ID — you can retrieve the job later via Job Retriever on the CHARMM-GUI homepage